

Hoi-Jun Yoo

professor

基本信息

姓名: Hoi-Jun Yoo

所属院校: 韩国科学技术院

地区: South Korea

年龄范围: 50岁以上

企业经历: 有

学科领域

电子与计算机工程

人工智能

集成电路设计与集成系统

生物医学工程

研究领域

低功耗设计与能耗优化

深度学习与神经网络

低功耗电路与能量管理

计算机体系结构

嵌入式系统与硬件设计

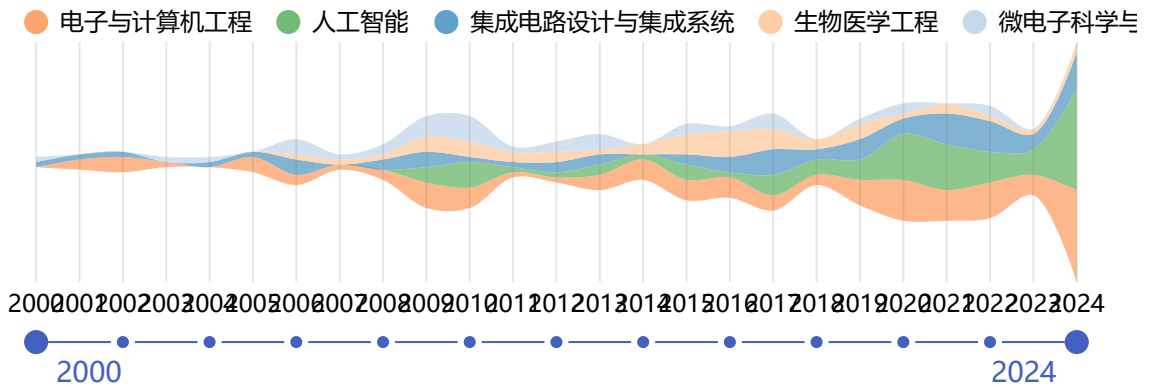
计算机视觉

生物医学传感与检测技术

个人简介

Hoi-Jun Yoo is a professor at the School of Electrical Engineering, KAIST. He holds a B.S., M.S., and Ph.D. in Electrical Engineering from Seoul National University and KAIST. Currently, he serves as the Dean of KAIST Graduate School of AI Semiconductor and holds multiple directorships. His research interests include AI Semiconductor, Artificial Intelligence System - on - Chip Design, etc. He has significant achievements like developing the world's first 256MB SDRAM. He has contributed to numerous research outputs, including 344 conference contributions, 225 articles, etc. His work also relates to UN Sustainable Development Goals.

近年发表的论文



A Low-Power Artificial-Intelligence-Based 3-D Rendering Processor With Hybrid Deep Neural Network Computing

IEEE MICRO

Han, Donghyeon; Ryu, Junha; Kim, Sangyeob; Kim, Sangjin; Park, Jongjun; Yoo, Hoi-Jun

被引用次数: 0

2024

COOL-NPU: Complementary Online Learning Neural Processing Unit

IEEE MICRO

Kim, Sangyeob; Kim, Soyeon; Hong, Seongyon; Kim, Sangjin; Choi, Jiwon; Han, Donghyeon; Yoo, Hoi-Jun

被引用次数: 0

2024

ED-MPIM: An Energy-Efficient Event-Driven Smart Vision SoC With High-Linearity and Reconfigurable MRAM PIM

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Xie, Wenao; Sang, Haoyang; Kwon, Beomseok; Im, Dongseok; Kim, Sangjin; Kim, Sangyeob; Lee, Kangho; Yoo, Hoi-Jun

被引用次数: 0

2024

NeuGPU: An Energy-Efficient Neural Graphics Processing Unit for Instant Modeling and Real-Time Rendering on Mobile Devices

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Ryu, Junha; Kwon, Hankyul; Park, Wonhoon; Li, Zhiyong; Kwon, Beomseok; Han, Donghyeon; Im, Dongseok; Kim, Sangyeob; Joo, Hyungnam; Kim, Minsung; Yoo, Hoi-Jun

被引用次数: 0

2024

LOG-CIM: An Energy-Efficient Logarithmic Quantization Computing-In-Memory Processor With Exponential Parallel Data Mapping and Zero-Aware 6T Dual-WL Cell

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Um, Soyeon; Kim, Sangjin; Hong, Seongyon; Kim, Sangyeob; Yoo, Hoi-Jun

被引用次数: 0

2024

A 3.8-mW 1.9-mΩ/√Hz Electrical Impedance Tomography IC With High Input Impedance and Loading Effect Calibration for 3-D Early Breast Cancer Detect System

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Um, Soyeon; Lee, Jaehyuk; Yoo, Hoi-Jun

被引用次数: 0

2024

Scaling-CIM: eDRAM In-Memory-Computing Accelerator With Dynamic-Scaling ADC and Adaptive Analog Operation

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Sangjin; Um, Soyeon; Jo, Wooyoung; Lee, Jingu; Ha, Sangwoo; Li, Zhiyong; Yoo, Hoi-Jun

被引用次数: 3

2024

C-DNN: An Energy-Efficient Complementary Deep-Neural-Network Processor With Heterogeneous CNN/SNN Core Architecture

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Sangyeob; Kim, Soyeon; Hong, Seongyon; Kim, Sangjin; Han, Donghyeon; Choi, Jiwon; Yoo, Hoi-Jun

被引用次数: 5

2024

DynaPlasia: An eDRAM In-Memory Computing-Based Reconfigurable Spatial Accelerator With Triple-Mode Cell

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Sangjin; Li, Zhiyong; Um, Soyeon; Jo, Wooyoung; Ha, Sangwoo; Lee, Juhyoung; Kim, Sangyeob; Han, Donghyeon; Yoo, Hoi-Jun

被引用次数: 7

2024

MetaVRain: A Mobile Neural 3-D Rendering Processor With Bundle-Frame-Familiarity-Based NeRF Acceleration and Hybrid DNN Computing

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Han, Donghyeon; Ryu, Junha; Kim, Sangyeob; Kim, Sangjin; Park, Jongjun; Yoo, Hoi-Jun

被引用次数: 4

2024

An Energy-Efficient CNN/Transformer Hybrid Neural Semantic Segmentation Processor With Chunk-Based Bit Plane Data Compression and Similarity-Based Token-Level Skipping Exploitation

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Park, Jongjun; Kim, Seryeong; Park, Wonhoon; Song, Seokchan; Yoo, Hoi-Jun

被引用次数: 0

2024

A 92 fps and 2.56 mJ/Frame Computing-In-Memory-Based Human Pose Estimation Accelerator With Resource-Efficient Macro for Mobile Devices

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Kwon, Beomseok; Li, Zhiyong; Kim, Sangjin; Jo, Wooyoung; Yoo, Hoi-Jun

被引用次数: 0

2024

A 8.81 TFLOPS/W Deep-Reinforcement-Learning Accelerator With Delta-Based Weight Sharing and Block-Mantissa Reconfigurable PE Array

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

An, Sanghyuk; Ryu, Junha; Park, Gwangtae; Yoo, Hoi-Jun

被引用次数: 0

2024

An 2.31uJ/Inference Ultra-Low Power Always-on Event-Driven AI-IoT SoC With Switchable nvSRAM Compute-in-Memory Macro

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Sang, Haoyang; Xie, Wenao; Park, Gwangtae; Yoo, Hoi-Jun

被引用次数: 1

2024

An Overview of Computing-in-Memory Circuits With DRAM and NVM

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Kim, Sangjin; Yoo, Hoi-Jun

被引用次数: 1

2024

Review of Non-invasive, Wearable Biomedical Sensor and Imaging IC

2024 IEEE BIOMEDICAL CIRCUITS AND SYSTEMS CONFERENCE, BIOCAS 2024

Jang, Jaeeun; Yoo, Hoi-Jun

被引用次数: 0

2024

Cache-PIM: An ECC-compatible eDRAM-PIM for Last-Level Cache with Resolution-aware Single-Cycle Voting

2024 50TH IEEE EUROPEAN SOLID-STATE ELECTRONICS RESEARCH CONFERENCE, ESSERC 2024

Ha, Sangwoo; Um, Soyeon; Kim, Sangjin; Sohn, Kyomin; Yoo, Hoi-Jun

被引用次数: 0

2024

A 28.6 mJ/iter Stable Diffusion Processor for Text-to-Image Generation with Patch Similarity-based Sparsity Augmentation and Text-based Mixed-Precision

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Choi, Jiwon; Jo, Wooyoung; Hong, Seongyon; Kwon, Beomseok; Park, Wonhoon; Yoo, Hoi-Jun

被引用次数: 0

2024

DualNet: Efficient Integration of Artificial Neural Network and Spiking Neural Network with Equivalent Conversion

2024 IEEE 6TH INTERNATIONAL CONFERENCE ON AI CIRCUITS AND SYSTEMS, AICAS 2024

Hong, Seongyon; Kim, Sangyeob; Kim, Soyeon; Yoo, Hoi-Jun

被引用次数: 0

2024

LUTein: Dense-Sparse Bit-slice Architecture with Radix-4 LUT-based Slice-Tensor Processing Units

2024 IEEE INTERNATIONAL SYMPOSIUM ON HIGH-PERFORMANCE COMPUTER ARCHITECTURE, HPCA 2024

Im, Dongseok; Yoo, Hoi-Jun

被引用次数: 0

2024

NoPIM: Functional Network-on-Chip Architecture for Scalable High-Density Processing-in-Memory-based Accelerator

2024 IEEE SYMPOSIUM IN LOW-POWER AND HIGH-SPEED CHIPS, COOL CHIPS 27

Kim, Sangjin; Li, Zhiyong; Um, Soyeon; Jo, Wooyoung; Ha, Sangwoo; Kim, Sangyeob; Yoo, Hoi-Jun

被引用次数: 0

2024

Two -Step Spike Encoding Scheme and Architecture for Highly Sparse Spiking -Neural -Network

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Kim, Sangyeob; Kim, Sangj In; Um, Soyeon; Kim, Soyeon; Yoo, Hoi-Jun

被引用次数: 0

2024

A Low-power and Real-time Neural-Rendering Dense SLAM Processor with 3-Level Hierarchical Sparsity Exploitation

2024 IEEE SYMPOSIUM IN LOW-POWER AND HIGH-SPEED CHIPS, COOL CHIPS 27

Park, Gwangtae; Song, Seokchan; Sang, Haoyang; Im, Dongseok; Han, Donghyeon; Kim, Sangyeob; Lee, Hongseok; Yoo, Hoi-Jun

被引用次数: 0

2024

A 3.55 mJ/frame Energy-efficient MixedTransformer based Semantic Segmentation Accelerator for Mobile Devices

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Park, Jongjun; Kim, Seryeong; Park, Wonhoon; Song, Seokchan; Yoo, Hoi-Jun

被引用次数: 0

2024

A Low-Power Neural Graphics System for Instant 3D Modeling and Real-Time Rendering on Mobile AR/VR Devices

2024 IEEE SYMPOSIUM IN LOW-POWER AND HIGH-SPEED CHIPS, COOL CHIPS 27

Ryu, Junha; Kwon, Hankyul; Park, Wonhoon; Li, Zhiyong; Kwon, Beomseok; Han, Donghyeon; Im, Dongseok; Kim, Sangyeob; Joo, Hyunnam; Kim, Minsung; Yoo, Hoi-Jun

被引用次数: 0

2024

CamPU: A Multi-Camera Processing Unit for Deep Learning-based 3D Spatial Computing Systems

IEEE/ACM International Symposium on Microarchitecture

Dongseok Im; Hoi-Jun Yoo

被引用次数: 0

2024

LUTein: Dense-Sparse Bit-Slice Architecture With Radix-4 LUT-Based Slice-Tensor Processing Units

IEEE International Symposium on High Performance Computer Architecture

Dongseok Im; Hoi-Jun Yoo

被引用次数: 0

2024

A Mobile 3-D Object Recognition Processor With Deep-Learning-Based Monocular Depth Estimation

IEEE MICRO

Im, Dongseok; Park, Gwangtae; Li, Zhiyong; Ryu, Junha; Kang, Sanghoon; Han, Donghyeon; Lee, Jinsu; Park, Wonhoon; Kwon, Hankyul; Yoo, Hoi-Jun

被引用次数: 0

2023

Neuro-CIM: ADC-Less Neuromorphic Computing-in-Memory Processor With Operation Gating/Stopping and Digital-Analog Networks

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Sangyeob; Kim, Sangjin; Um, Soyeon; Kim, Soyeon; Kim, Kwantae; Yoo, Hoi-Jun

被引用次数: 8

2023

SNPU: An Energy-Efficient Spike Domain Deep-Neural-Network Processor With Two-Step Spike Encoding and Shift-and-Accumulation Unit

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Sangyeob; Kim, Sangjin; Um, Soyeon; Kim, Soyeon; Lee, Juhyoung; Yoo, Hoi-Jun

被引用次数: 3

2023

An Efficient Deep-Learning-Based Super-Resolution Accelerating SoC With Heterogeneous Accelerating and Hierarchical Cache

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Li, Zhiyong; Kim, Sangjin; Im, Dongseok; Han, Donghyeon; Yoo, Hoi-Jun

被引用次数: 6

2023

C-DNN V2: Complementary Deep-Neural-Network Processor With Full-Adder/OR-Based Reduction Tree and Reconfigurable Spatial Weight Reuse

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Kim, Sangyeob; Yoo, Hoi-Jun

被引用次数: 1

2023

A 3.8 mW 1.9 m Ω /√Hz Electrical Impedance Tomography Imaging with 28.4 M Ω High Input Impedance and Loading Calibration

IEEE 49TH EUROPEAN SOLID STATE CIRCUITS CONFERENCE, ESSCIRC 2023

Um, Soyeon; Lee, Jaehyuk; Yoo, Hoi-Jun

被引用次数: 2

2023

Sibia: Signed Bit-slice Architecture for Dense DNN Acceleration with Slice-level Sparsity Exploitation

IEEE International Symposium on High Performance Computer Architecture

Dongseok Im; Gwangtae Park; Zhiyong Li; Junha Ryu; Hoi-Jun Yoo

被引用次数: 0

2023

A Mobile DNN Training Processor With Automatic Bit Precision Search and Fine-Grained Sparsity Exploitation

IEEE MICRO

Han, Donghyeon; Im, Dongseok; Park, Gwangtae; Kim, Youngwoo; Song, Seokchan; Lee, Juhyoung; Yoo, Hoi-Jun

被引用次数: 0

2022

ECIM: Exponent Computing in Memory for an Energy-Efficient Heterogeneous Floating-Point DNN Training Processor

IEEE MICRO

Lee, Juhyoung; Kim, Jihoon; Jo, Wooyoung; Kim, Sangyeob; Kim, Sangjin; Yoo, Hoi-Jun

被引用次数: 8

2022

DSPU: An Efficient Deep Learning-Based Dense RGB-D Data Acquisition With Sensor Fusion and 3-D Perception SoC

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Im, Dongseok; Park, Gwangtae; Ryu, Junha; Li, Zhiyong; Kang, Sanghoon; Han, Donghyeon; Lee, Jinsu; Park, Wonhoon; Kwon, Hankyul; Yoo, Hoi-Jun

被引用次数: 3

2022

A 23- μ W Keyword Spotting IC With Ring-Oscillator-Based Time-Domain Feature Extraction

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Kwantae; Gao, Chang; Graca, Rui; Kiselev, Ilya; Yoo, Hoi-Jun; Delbruck, Tobi; Liu, Shih-Chii

被引用次数: 14

2022

FlashMAC: A Time-Frequency Hybrid MAC Architecture With Variable Latency-Aware Scheduling for TinyML Systems

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Gweon, Surin; Kang, Sanghoon; Kim, Kwantae; Yoo, Hoi-Jun

被引用次数: 3

2022

OmniDRL: An Energy-Efficient Deep Reinforcement Learning Processor With Dual-Mode Weight Compression and Sparse Weight Transposer

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lee, Juhyoung; Kim, Sangyeob; Kim, Sangjin; Jo, Wooyoung; Kim, Ji-Hoon; Han, Donghyeon; Yoo, Hoi-Jun

被引用次数: 2

2022

Design of Sub-10- μ W Sub-0.1% THD Sinusoidal Current Generator IC for Bio-Impedance Sensing

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Kwantae; Kim, Sangyeob; Yoo, Hoi-Jun

被引用次数: 16

2022

A Pipelined Point Cloud Based Neural Network Processor for 3-D Vision With Large-Scale Max Pooling Layer Prediction

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Im, Dongseok; Han, Donghyeon; Kang, Sanghoon; Yoo, Hoi-Jun

被引用次数: 7

2022

A Low-Power Graph Convolutional Network Processor With Sparse Grouping for 3D Point Cloud Semantic Segmentation in Mobile Devices

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Kim, Sangjin; Kim, Sangyeob; Lee, Juhyoung; Yoo, Hoi-Jun

被引用次数: 7

2022

TSUNAMI: Triple Sparsity-Aware Ultra Energy-Efficient Neural Network Training Accelerator With Multi-Modal Iterative Pruning

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Kim, Sangyeob; Lee, Juhyoung; Kang, Sanghoon; Han, Donghyeon; Jo, Wooyoung; Yoo, Hoi-Jun

被引用次数: 10

2022

A 36.2 dB High SNR and PVT/Leakage-Robust eDRAM Computing-In-Memory Macro With Segmented BL and Reference Cell Array

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Ha, Sangwoo; Kim, Sangjin; Han, Donghyeon; Um, Soyeon; Yoo, Hoi-Jun

被引用次数: 16

2022

A 49.5 mW Multi-Scale Linear Quantized Online Learning Processor for Real-Time Adaptive Object Detection

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Song, Seokchan; Kim, Soyeon; Park, Gwangtae; Han, Donghyeon; Yoo, Hoi-Jun

被引用次数: 1

2022

An Energy-Efficient GAN Accelerator With On-Chip Training for Domain-Specific Optimization

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Soyeon; Kang, Sanghoon; Han, Donghyeon; Kim, Sangjin; Kim, Sangyeob; Yoo, Hoi-Jun

被引用次数: 4

2021

GANPU: An Energy-Efficient Multi-DNN Training Processor for GANs With Speculative Dual-Sparsity Exploitation

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kang, Sanghoon; Han, Donghyeon; Lee, Juhyoung; Im, Dongseok; Kim, Sangyeob; Kim, Soyeon; Ryu, Junha; Yoo, Hoi-Jun

被引用次数: 23

2021

HNPU: An Adaptive DNN Training Processor Utilizing Stochastic Dynamic Fixed-Point and Active Bit-Precision Searching

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Han, Donghyeon; Im, Dongseok; Park, Gwangtae; Kim, Youngwoo; Song, Seokchan; Lee, Juhyoung; Yoo, Hoi-Jun

被引用次数: 35

2021

A 36-Channel Auto-Calibrated Front-End ASIC for a pMUT-Based Miniaturized 3-D Ultrasound System

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lee, Jihee; Lee, Kyoung-Rog; Eovino, Benjamin E.; Park, Jeong Hoan; Liang, Luna Yue; Lin, Liwei; Yoo, Hoi-Jun; Yoo, Jerald

被引用次数: 29

2021

DF-LNPU: A Pipelined Direct Feedback Alignment-Based Deep Neural Network Learning Processor for Fast Online Learning

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Han, Donghyeon; Lee, Jinsu; Yoo, Hoi-Jun

被引用次数: 10

2021

A 9.6-mW/Ch 10-MHz Wide-Bandwidth Electrical Impedance Tomography IC With Accurate Phase Compensation for Early Breast Cancer Detection

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lee, Jaehyuk; Gweon, Surin; Lee, Kwonjoon; Um, Soyeon; Lee, Kyoung-Rog; Yoo, Hoi-Jun

被引用次数: 27

2021

A 43.1TOPS/W Energy-Efficient Absolute-Difference-Accumulation Operation Computing-In-Memory With Computation Reuse

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Um, Soyeon; Kim, Sangyeob; Kim, Sangjin; Yoo, Hoi-Jun

被引用次数: 5

2021

A 64.1mW Accurate Real-Time Visual Object Tracking Processor With Spatial Early Stopping on Siamese Network

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Kim, Soyeon; Kim, Sangjin; Kim, Sangyeob; Han, Donghyeon; Yoo, Hoi-Jun

被引用次数: 7

2021

A 0.82 μ W CIS-Based Action Recognition SoC With Self-Adjustable Frame Resolution for Always-on IoT Devices

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Ryu, Junha; Park, Gwangtae; Im, Dongseok; Kim, Ji-Hoon; Yoo, Hoi-Jun

被引用次数: 1

2021

An Overview of Sparsity Exploitation in CNNs for On-Device Intelligence With Software-Hardware Cross-Layer Optimizations

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Kang, Sanghoon; Park, Gwangtae; Kim, Sangjin; Kim, Soyeon; Han, Donghyeon; Yoo, Hoi-Jun

被引用次数: 12

2021

Simultaneous Electrical Bio-Impedance Plethysmography at Different Body Parts: Continuous and Non-Invasive Monitoring of Pulse Wave Velocity

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Lee, Kwonjoon; Yoo, Hoi-Jun

被引用次数: 14

2021

AI SoCs for AR/VR User-Interaction

2021 IEEE INTERNATIONAL ELECTRON DEVICES MEETING (IEDM)

Ryu, Junha; Im, Dongseok; Yoo, Hoi-Jun

被引用次数: 1

2021

FlashMAC: An Energy-Efficient Analog-Digital Hybrid MAC with Variable Latency-Aware Scheduling

IEEE ASIAN SOLID-STATE CIRCUITS CONFERENCE (A-SSCC 2021)

Gweon, Surin; Kang, Sanghoon; Han, Donghyeon; Lee, Kyoung-Rog; Kim, Kwantae; Yoo, Hoi-Jun

被引用次数: 0

2021

An Energy-Efficient Deep Reinforcement Learning FPGA Accelerator for Online Fast Adaptation with Selective Mixed-precision Re-training

IEEE ASIAN SOLID-STATE CIRCUITS CONFERENCE (A-SSCC 2021)

Jo, Wooyoung; Lee, Juhyoung; Park, Seunghyun; Yoo, Hoi-Jun

被引用次数: 2

2021

Energy-Efficient Deep Reinforcement Learning Accelerator Designs for Mobile Autonomous Systems

2021 IEEE 3RD INTERNATIONAL CONFERENCE ON ARTIFICIAL INTELLIGENCE CIRCUITS AND SYSTEMS (AICAS)

Lee, Juhyoung; Kim, Changhyeon; Han, Donghyeon; Kim, Sangyeob; Kim, Sangjin; Yoo, Hoi-Jun

被引用次数: 2

2021

Neuro-inspired computing chips

NATURE ELECTRONICS

Zhang, Wenqiang; Gao, Bin; Tang, Jianshi; Yao, Peng; Yu, Shimeng; Chang, Meng-Fan; Yoo, Hoi-Jun; Qian, He; Wu, Huaqiang

被引用次数: 443

2020

The Heterogeneous Deep Neural Network Processor With a Non-von Neumann Architecture

PROCEEDINGS OF THE IEEE

Shin, Dongjoo; Yoo, Hoi-Jun

被引用次数: 29

2020

Wireless Body-Area-Network Transceiver and Low-Power Receiver With High Application Expandability

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jang, Jaeeun; Lee, Jihee; Cho, Hyunwoo; Lee, Jaehyuk; Yoo, Hoi-Jun

被引用次数: 15

2020

A 0.5-V Sub-10- μ W 15.28-m Ω /√Hz Bio-Impedance Sensor IC With Sub-1° Phase Error

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Kwantae; Kim, Ji-Hoon; Gweon, Surin; Kim, Minseo; Yoo, Hoi-Jun

被引用次数: 23

2020

The Development of Silicon for AI: Different Design Approaches

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Lee, Kyuho Jason; Lee, Jinmook; Choi, Sungpill; Yoo, Hoi-Jun

被引用次数: 10

2020

The Hardware and Algorithm Co-Design for Energy-Efficient DNN Processor on Edge/Mobile Devices

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Lee, Jinsu; Kang, Sanghoon; Lee, Jinmook; Shin, Dongjoo; Han, Donghyeon; Yoo, Hoi-Jun

被引用次数: 29

2020

DT-CNN: An Energy-Efficient Dilated and Transposed Convolutional Neural Network Processor for Region of Interest Based Image Segmentation

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Im, Dongseok; Han, Donghyeon; Choi, Sungpill; Kang, Sanghoon; Yoo, Hoi-Jun

被引用次数: 26

2020

A Power-Efficient CNN Accelerator With Similar Feature Skipping for Face Recognition in Mobile Devices

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Kim, Sangyeob; Lee, Juhyoung; Kang, Sanghoon; Lee, Jinsu; Yoo, Hoi-Jun

被引用次数: 18

2020

A 1.15 TOPS/W Energy-Efficient Capsule Network Accelerator for Real-Time 3D Point Cloud Segmentation in Mobile Environment

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Park, Gwangtae; Im, Dongseok; Han, Donghyeon; Yoo, Hoi-Jun

被引用次数: 11

2020

A 0.22-0.89 mW Low-Power and Highly-Secure Always-On Face Recognition Processor With Adversarial Attack Prevention

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Kim, Youngwoo; Han, Donghyeon; Kim, Changhyeon; Yoo, Hoi-Jun

被引用次数: 8

2020

SRNPU: An Energy-Efficient CNN-Based Super-Resolution Processor With Tile-Based Selective Super-Resolution in Mobile Devices

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Lee, Juhyoung; Lee, Jinsu; Yoo, Hoi-Jun

被引用次数: 36

2020

An Ultra-low-power Mixed-mode Face Recognition Processor for Always-on User Authentication in Mobile Device

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Kim, Ji-Hoon; Kim, Changhyeon; Kim, Kwantae; Lee, Juhyoung; Yoo, Hoi-Jun; Kim, Joo-Young

被引用次数: 2

2020

A 9.52 ms Latency, and Low-power Streaming Depth-estimation Processor with Shifter-based Pipelined Architecture for Smart Mobile Devices

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Choi, Sungpill; Lee, Kyuho Jason; Kim, Youngwoo; Yoo, Hoi-Jun

被引用次数: 0

2020

A 0.8-V 82.9- μ W In-Ear BCI Controller IC With 8.8 PEF EEG Instrumentation Amplifier and Wireless BAN Transceiver

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lee, Jaehyuk; Lee, Kyoung-Rog; Ha, Unsoo; Kim, Ji-Hoon; Lee, Kwonjoon; Gweon, Surin; Jang, Jaeeun; Yoo, Hoi-Jun

被引用次数: 33

2019

A Four-Camera VGA-Resolution Capsule Endoscope System With 80-Mb/s Body Channel Communication Transceiver and Sub-Centimeter Range Capsule Localization

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jang, Jaeeun; Lee, Jihee; Lee, Kyoung-Rog; Lee, Jiwon; Kim, Minseo; Lee, Yongsu; Bae, Joonsung; Yoo, Hoi-Jun

被引用次数: 35

2019

UNPU: An Energy-Efficient Deep Neural Network Accelerator With Fully Variable Weight Bit Precision

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lee, Jinmook; Kim, Changhyeon; Kang, Sanghoon; Shin, Dongjoo; Kim, Sangyeob; Yoo, Hoi-Jun

被引用次数: 240

2019

A Low-Power Deep Neural Network Online Learning Processor for Real-Time Object Tracking Application

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Han, Donghyeon; Lee, Jinsu; Lee, Jinmook; Yoo, Hoi-Jun

被引用次数: 34

2019

Custom Sub-Systems and Circuits for Deep Learning: Guest Editorial Overview

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Chen, Chia-Yu; Murmann, Boris; Seo, Jae-Sun; Yoo, Hoi-Jun

被引用次数: 2

2019

Low Power and Accurate Current Driver IC for Electrical Impedance Tomography Applications

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Jang, Jaeeun; Yoo, Hoi-Jun

被引用次数: 2

2019

A 99.4 fps Optical Flow Estimation Processor with Image Tiling for Action Recognition in Mobile Devices

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Lee, Juhyoung; Choi, Sungpill; Lee, Jinmook; Kang, Sanghoon; Yoo, Hoi-Jun

被引用次数: 0

2019

A Chaotic Entropy Source based PUF Design for Secure IoT Platform

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Kim, Minseo; Yoo, Hoi-Jun

被引用次数: 0

2019

A Low-power, Mixed-mode Neural Network Classifier for Robust Scene Classification

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Lee, Kyuho; Park, Junyoung; Yoo, Hoi-Jun

被引用次数: 23

2019

The Reinforcement Learning based Local Routing Optimization for Ad-hoc Network

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Lee, Yongsu; Woo, Jongchan; Yoo, Hoi-Jun

被引用次数: 0

2019

Direct Feedback Alignment Based Convolutional Neural Network Training for Low-Power Online Learning Processor

International Conference on Computer Vision

Donghyeon Han; Hoi-jun Yoo

被引用次数: 0

2019

DNPU: An Energy-Efficient Deep-Learning Processor with Heterogeneous Multi-Core Architecture

IEEE MICRO

Shin, Dongjoo; Lee, Jinmook; Lee, Jinsu; Lee, Juhyoung; Yoo, Hoi-Jun

被引用次数: 50

2018

An EEG-NIRS Multimodal SoC for Accurate Anesthesia Depth Monitoring

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Ha, Unsoo; Lee, Jaehyuk; Kim, Minseo; Roh, Taehwan; Choi, Sangsik; Yoo, Hoi-Jun

被引用次数: 61

2018

A Low-Power Convolutional Neural Network Face Recognition Processor and a CIS Integrated With Always-on Face Detector

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Bong, Kyeongryeol; Choi, Sungpill; Kim, Changhyeon; Han, Donghyeon; Yoo, Hoi-Jun

被引用次数: 87

2018

A 0.55 V 1.1 mW Artificial Intelligence Processor With On-Chip PVT Compensation for Autonomous Mobile Robots

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Kim, Youchang; Shin, Dongjoo; Lee, Jinsu; Lee, Yongsu; Yoo, Hoi-Jun

被引用次数: 10

2018

Low-Power Scalable 3-D Face Frontalization Processor for CNN-Based Face Recognition in Mobile Devices

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Kang, Sanghoon; Lee, Jinmook; Bong, Kyeongryeol; Kim, Changhyeon; Kim, Youchang; Yoo, Hoi-Jun

被引用次数: 5

2018

Toward all-day wearable health monitoring: An ultralow-power, reflective organic pulse oximetry sensing patch

SCIENCE ADVANCES

Lee, Hyeonwoo; Kim, Eunhye; Lee, Yongsu; Kim, Hoyeon; Lee, Jaeho; Kim, Mincheol; Yoo, Hoi-Jun; Yoo, Seunghyup

被引用次数: 205

2018

Low-Power Convolutional Neural Network Processor for a Face-Recognition System

IEEE MICRO

Bong, Kyeongryeol; Choi, Sungpill; Kim, Changhyeon; Yoo, Hoi-Jun

被引用次数: 23

2017

BRAIN: A Low-Power Deep Search Engine for Autonomous Robots

IEEE MICRO

Kim, Youchang; Shin, Dongjoo; Lee, Jinsu; Yoo, Hoi-Jun

被引用次数: 5

2017

A 1.4-m Ω -Sensitivity 94-dB Dynamic-Range Electrical Impedance Tomography SoC and 48-Channel Hub-SoC for 3-D Lung Ventilation Monitoring System

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Minseo; Jang, Jaeun; Kim, Hyunki; Lee, Jihee; Lee, Jaehyuck; Lee, Jiwon; Lee, Kyoung-Rog; Kim, Kwantae; Lee, Yongsu; Lee, Kyuho Jason; Yoo, Hoi-Jun

被引用次数: 50

2017

A 82-nW Chaotic Map True Random Number Generator Based on a Sub-Ranging SAR ADC

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Minseo; Ha, Unsoo; Lee, Kyuho Jason; Lee, Yongsu; Yoo, Hoi-Jun

被引用次数: 73

2017

A 502-GOPS and 0.984-mW Dual-Mode Intelligent ADAS SoC With Real-Time Semiglobal Matching and Intention Prediction for Smart Automotive Black Box System

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lee, Kyuho Jason; Bong, Kyeongryeol; Kim, Changhyeon; Jang, Jaeun; Lee, Kyoung-Rog; Lee, Jihee; Kim, Gyeonghoon; Yoo, Hoi-Jun

被引用次数: 32

2017

An Energy-Efficient Speech-Extraction Processor for Robust User Speech Recognition in Mobile Head-Mounted Display Systems

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Lee, Jinmook; Park, Seongwook; Hong, Injoon; Yoo, Hoi-Jun

被引用次数: 5

2017

Sticker-Type Hybrid Photoplethysmogram Monitoring System Integrating CMOS IC With Organic Optical Sensors

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Lee, Yongsu; Lee, Hyeonwoo; Jang, Jaeeun; Lee, Jihee; Kim, Minseo; Lee, Jaehyuk; Kim, Hyunki; Yoo, Seunghyup; Yoo, Hoi-Jun

被引用次数: 30

2017

A 17.5-fJ/bit Energy-Efficient Analog SRAM for Mixed-Signal Processing

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Lee, Jinsu; Shin, Dongjoo; Kim, Youchang; Yoo, Hoi-Jun

被引用次数: 21

2017

A 206.3 μ W Non-contact Compensation IC for Body Channel Communication

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Lee, Kyoung-Rog; Jang, Jaeeun; Yoo, Hoi-Jun

被引用次数: 0

2017

A 55.77 μ W Bio-impedance Sensor with 276 μ s Settling Time for Portable Blood Pressure Monitoring System

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Kim, Kwantae; Kim, Minseo; Cho, Hyunwoo; Lee, Kwonjoon; Yoo, Hoi-Jun

被引用次数: 2

2017

0.025 mJ/image Fast-scan and SNR Enhanced Electrical Impedance Tomography IC for Lung Ventilation Monitoring

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Lee, Jaehyuk; Ha, Unsoo; Yoo, Hoi-Jun

被引用次数: 3

2017

A 48 μ W, 8.88 mW/W Batteryless Wirelessly Power Transferred Body Channel Communication Identification System

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Lee, Jihee; Lee, Yongsu; Cho, Hyunwoo; Yoo, Hoi-Jun

被引用次数: 0

2017

80 μ W/MHz, 850 MHz Fault Tolerant Processor with Fault Monitor Systems

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Han, Jinho; Kwon, Youngsu; Shin, Kyeongsun; Yoo, Hoi-Jun

被引用次数: 1

2017

A Memory-efficient Hand Segmentation Architecture for Hand Gesture Recognition in Low-power Mobile Devices

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Choi, Sungpill; Park, Seongwook; Yoo, Hoi-Jun

被引用次数: 1

2017

An Energy-Efficient Embedded Deep Neural Network Processor for High Speed Visual Attention in Mobile Vision Recognition SoC

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Park, Seongwook; Hong, Injoon; Park, Junyoung; Yoo, Hoi-Jun

被引用次数: 21

2016

A 0.5° Error 10 mW CMOS Image Sensor-Based Gaze Estimation Processor

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Bong, Kyeongryeol; Hong, Injoon; Kim, Gyeonghoon; Yoo, Hoi-Jun

被引用次数: 10

2016

A 2.71 nJ/Pixel Gaze-Activated Object Recognition System for Low-Power Mobile Smart Glasses

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Hong, Injoon; Bong, Kyeongryeol; Shin, Dongjoo; Park, Seongwook; Lee, Kyuho Jason; Kim, Youchang; Yoo, Hoi-Jun

被引用次数: 23

2016

A 79 pJ/b 80 Mb/s Full-Duplex Transceiver and a 42.5μW 100 kb/s Super-Regenerative Transceiver for Body Channel Communication

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Cho, Hyunwoo; Kim, Hyunki; Kim, Minseo; Jang, Jaeeun; Lee, Yongsu; Lee, Kyuho Jason; Bae, Joonsung; Yoo, Hoi-Jun

被引用次数: 68

2016

A CMOS Image Sensor-Based Stereo Matching Accelerator With Focal-Plane Sparse Rectification and Analog Census Transform

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Kim, Changhyeon; Bong, Kyeongryeol; Choi, Sungpill; Lee, Kyuho Jason; Yoo, Hoi-Jun

被引用次数: 5

2016

A 0.5 V 54 μW Ultra-Low-Power Object Matching Processor for Micro Air Vehicle Navigation

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Kim, Youchang; Hong, Injoon; Park, Junyoung; Yoo, Hoi-Jun

被引用次数: 9

2016

A Fault-Tolerant Cache System of Automotive Vision Processor Complying With ISO26262

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Han, Jinho; Kwon, Youngsu; Byun, Kyeongjin; Yoo, Hoi-Jun

被引用次数: 10

2016

The effects of electrode impedance on receiver sensitivity in body channel communication

MICROELECTRONICS JOURNAL

Bae, Joonsung; Yoo, Hoi-Jun

被引用次数: 5

2016

A Wrist Watch-type Cardiovascular Monitoring System using Concurrent ECG and APW Measurement

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Lee, Kwonjoon; Song, Kiseok; Roh, Taehwan; Yoo, Hoi-jun

被引用次数: 3

2016

A 95% accurate EEG-connectome Processor for a Mental Health Monitoring System

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Kim, Hyunki; Song, Kiseok; Roh, Taehwan; Yoo, Hoi-Jun

被引用次数: 2

2016

A 540- μ W Duty Controlled RSSI With Current Reusing Technique for Human Body Communication

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Jang, Jaeun; Lee, Yongsu; Cho, Hyunwoo; Yoo, Hoi-Jun

被引用次数: 6

2016

A 259.6 μ W HRV-EEG Processor With Nonlinear Chaotic Analysis During Mental Tasks

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Roh, Taehwan; Hong, Sunjoo; Cho, Hyunwoo; Yoo, Hoi-Jun

被引用次数: 11

2016

A 10.4 mW Electrical Impedance Tomography SoC for Portable Real-Time Lung Ventilation Monitoring System

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Hong, Sunjoo; Lee, Jaehyuk; Bae, Joonsung; Yoo, Hoi-Jun

被引用次数: 62

2015

A 27 mW Reconfigurable Marker-Less Logarithmic Camera Pose Estimation Engine for Mobile Augmented Reality Processor

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Hong, Injoon; Kim, Gyeonghoon; Kim, Youchang; Kim, Donghyun; Nam, Byeong-Gyu; Yoo, Hoi-Jun

被引用次数: 15

2015

A 5.2 mW IEEE 802.15.6 HBC Standard Compatible Transceiver With Power Efficient Delay-Locked-Loop Based BPSK Demodulator

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Cho, Hyunwoo; Lee, Hyungwoo; Bae, Joonsung; Yoo, Hoi-Jun

被引用次数: 27

2015

A 45 μ W Injection-Locked FSK Wake-Up Receiver With Frequency-to-Envelope Conversion for Crystal-Less Wireless Body Area Network

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Bae, Joonsung; Yoo, Hoi-Jun

被引用次数: 42

2015

An Impedance and Multi-Wavelength Near-Infrared Spectroscopy IC for Non-Invasive Blood Glucose Estimation

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Song, Kiseok; Ha, Unsoo; Park, Seongwook; Bae, Joonsung; Yoo, Hoi-Jun

被引用次数: 62

2015

A Vocabulary Forest Object Matching Processor With 2.07 M-Vector/s Throughput and 13.3 nJ/Vector Per-Vector Energy for Full-HD 60 fps Video Object Recognition

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lee, Kyuho Jason; Kim, Gyeonghoon; Park, Junyoung; Yoo, Hoi-Jun

被引用次数: 17

2015

A 1.22 TOPS and 1.52 mW/MHz Augmented Reality Multicore Processor With Neural Network NoC for HMD Applications

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Gyeonghoon; Lee, Kyuho; Kim, Youchang; Park, Seongwook; Hong, Injoon; Bong, Kyeongryeol; Yoo, Hoi-Jun

被引用次数: 22

2015

A 4.9 mΩ-Sensitivity Mobile Electrical Impedance Tomography IC for Early Breast-Cancer Detection System

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Hong, Sunjoo; Lee, Kwonjoon; Ha, Unsoo; Kim, Hyunki; Lee, Yongsu; Kim, Youchang; Yoo, Hoi-Jun

被引用次数: 119

2015

The Effects of Electrode Configuration on Body Channel Communication Based on Analysis of Vertical and Horizontal Electric Dipoles

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Bae, Joonsung; Yoo, Hoi-Jun

被引用次数: 33

2015

A Wearable EEG-HEG-HRV Multimodal System With Simultaneous Monitoring of tES for Mental Health Management

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Ha, Unsoo; Lee, Yongsu; Kim, Hyunki; Roh, Taehwan; Bae, Joonsung; Kim, Changhyeon; Yoo, Hoi-Jun

被引用次数: 42

2015

An Energy-Efficient and Scalable Deep Learning/Inference Processor With Tetra-Parallel MIMD Architecture for Big Data Applications

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Park, Seong-Wook; Park, Junyoung; Bong, Kyeongryeol; Shin, Dongjoo; Lee, Jinmook; Choi, Sungpill; Yoo, Hoi-Jun

被引用次数: 31

2015

AN AUGMENTED REALITY PROCESSOR WITH A CONGESTION-AWARE NETWORK-ON-CHIP SCHEDULER

IEEE MICRO

Kim, Gyeonghoon; Kim, Donghyun; Park, Seongwook; Kim, Youchang; Lee, Kyuho; Hong, Injoon; Bong, Kyeongryeol; Yoo, Hoi-Jun

被引用次数: 4

2014

An 87-mA . Iontophoresis Controller IC With Dual-Mode Impedance Sensor for Patch-Type Transdermal Drug Delivery System

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Song, Kiseok; Ha, Unsoo; Lee, Jaehyuk; Bong, Kyeongryeol; Yoo, Hoi-Jun

被引用次数: 17

2014

Intelligent Network-on-Chip With Online Reinforcement Learning for Portable HD Object Recognition Processor

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Park, Junyoung; Hong, Injoon; Kim, Gyeonghoon; Nam, Byeong-Gyu; Yoo, Hoi-Jun

被引用次数: 9

2014

Hybrid radix-4/-8 truncated multiplier for mobile GPU applications

ELECTRONICS LETTERS

Choi, Seongrim; Kim, Gyeonghoon; Yoo, Hoi-Jun; Nam, Byeong-Gyu

被引用次数: 2

2014

A Wearable Neuro-Feedback System With EEG-Based Mental Status Monitoring and Transcranial Electrical Stimulation

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Roh, Taehwan; Song, Kiseok; Cho, Hyunwoo; Shin, Dongjoo; Yoo, Hoi-Jun

被引用次数: 38

2014

A 57 mW 12.5 μ J/Epoch Embedded Mixed-Mode Neuro-Fuzzy Processor for Mobile Real-Time Object Recognition

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Oh, Jinwook; Kim, Gyeonghoon; Nam, Byeong-Gyu; Yoo, Hoi-Jun

被引用次数: 17

2013

An 86 mW 98GOPS ANN-Searching Processor for Full-HD 30 fps Video Object Recognition With Zeroless Locality-Sensitive Hashing

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Gyeonghoon; Oh, Jinwook; Lee, Seungjin; Yoo, Hoi-Jun

被引用次数: 13

2013

A 320 mW 342 GOPS Real-Time Dynamic Object Recognition Processor for HD 720p Video Streams

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Oh, Jinwook; Kim, Gyeonghoon; Park, Junyoung; Hong, Injoon; Lee, Seungjin; Kim, Joo-Young; Woo, Jeong-Ho; Yoo, Hoi-Jun

被引用次数: 24

2013

A 37.5 μ W Body Channel Communication Wake-Up Receiver With Injection-Locking Ring Oscillator for Wireless Body Area Network

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Cho, Hyunwoo; Bae, Joonsung; Yoo, Hoi-Jun

被引用次数: 52

2013

1.2-mW Online Learning Mixed-Mode Intelligent Inference Engine for Low-Power Real-Time Object Recognition Processor

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Oh, Jinwook; Lee, Seungjin; Yoo, Hoi-Jun

被引用次数: 13

2013

Low Power and Self-Reconfigurable WBAN Controller for Continuous Bio-Signal Monitoring System

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Lee, Seulki; Yoo, Hoi-Jun

被引用次数: 6

2013

LOW-POWER, REAL-TIME OBJECT-RECOGNITION PROCESSORS FOR MOBILE VISION SYSTEMS

IEEE MICRO

Oh, Jinwook; Kim, Gyeonghoon; Hong, Injoon; Park, Junyoung; Lee, Seungjin; Kim, Joo-Young; Woo, Jeong-Ho; Yoo, Hoi-Jun

被引用次数: 9

2012

A Low-Energy Crystal-Less Double-FSK Sensor Node Transceiver for Wireless Body-Area Network

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Bae, Joonsung; Song, Kiseok; Lee, Hyungwoo; Cho, Hyunwoo; Yoo, Hoi-Jun

被引用次数: 57

2012

A 92-mW Real-Time Traffic Sign Recognition System With Robust Illumination Adaptation and Support Vector Machine

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Park, Junyoung; Kwon, Joonsoo; Oh, Jinwook; Lee, Seungjin; Kim, Joo-Young; Yoo, Hoi-Jun

被引用次数: 10

2012

A 0.24-nJ/b Wireless Body-Area-Network Transceiver With Scalable Double-FSK Modulation

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Bae, Joonsung; Song, Kiseok; Lee, Hyungwoo; Cho, Hyunwoo; Yoo, Hoi-Jun

被引用次数: 99

2012

A 75 μ W Real-Time Scalable Body Area Network Controller and a 25 μ W ExG Sensor IC for Compact Sleep Monitoring Applications

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lee, Seulki; Yan, Long; Roh, Taehwan; Hong, Sunjoo; Yoo, Hoi-Jun

被引用次数: 21

2012

The Signal Transmission Mechanism on the Surface of Human Body for Body Channel Communication

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Bae, Joonsung; Cho, Hyunwoo; Song, Kiseok; Lee, Hyungwoo; Yoo, Hoi-Jun

被引用次数: 220

2012

Emerging Circuits and Systems Techniques for Ultra-Low Power Body Sensor Networks

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Calhoun, Benton; Chandrakasan, Anantha P.; Otis, Brian; Verma, Naveen; Yoo, Hoi-Jun

被引用次数: 0

2012

A Sub-10 nA DC-Balanced Adaptive Stimulator IC With Multi-Modal Sensor for Compact Electro-Acupuncture Stimulation

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Song, Kiseok; Lee, Hyungwoo; Hong, Sunjoo; Cho, Hyunwoo; Ha, Unsoo; Yoo, Hoi-Jun

被引用次数: 18

2012

A Low Energy Injection-Locked FSK Transceiver With Frequency-to-Amplitude Conversion for Body Sensor Applications

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Bae, Joonsung; Yan, Long; Yoo, Hoi-Jun

被引用次数: 112

2011

A 345 mW Heterogeneous Many-Core Processor With an Intelligent Inference Engine for Robust Object Recognition

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lee, Seungjin; Oh, Jinwook; Park, Junyoung; Kwon, Joonsoo; Kim, Minsu; Yoo, Hoi-Jun

被引用次数: 42

2011

A 3.9 mW 25-Electrode Reconfigured Sensor for Wearable Cardiac Monitoring System

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Yan, Long; Bae, Joonsung; Lee, Seulki; Roh, Taehwan; Song, Kiseok; Yoo, Hoi-Jun

被引用次数: 102

2011

A 40fJ/c-s 1 V 10 bit SARADC with Dual Sampling Capacitive DAC Topology

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Kim, Binhee; Yan, Long; Yoo, Jerald; Yoo, Hoi-Jun

被引用次数: 5

2011

A Wirelessly Powered Electro-Acupuncture Based on Adaptive Pulsewidth Monophase Stimulation

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Song, Kiseok; Yan, Long; Lee, Seulki; Yoo, Jerald; Yoo, Hoi-Jun

被引用次数: 10

2011

Familiarity based unified visual attention model for fast and robust object recognition

PATTERN RECOGNITION

Lee, Seungjin; Kim, Kwanho; Kim, Joo-Young; Kim, Minsu; Yoo, Hoi-Jun

被引用次数: 37

2010

A 4.78 μ s Dynamic Compensated Inductive Coupling Transceiver for Ubiquitous and Wearable Body Sensor Network

IEICE TRANSACTIONS ON COMMUNICATIONS

Lee, Seulki; Yoo, Jerald; Yoo, Hoi-Jun

被引用次数: 0

2010

Visual Image Processing RAM: Memory Architecture With 2-D Data Location Search and Data Consistency Management for a Multicore Object Recognition Processor

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY

Kim, Joo-Young; Kim, Donghyun; Lee, Seungjin; Kim, Kwanho; Yoo, Hoi-Jun

被引用次数: 5

2010

A 0.5- μ Vrms 12- μ W Wirelessly Powered Patch-Type Healthcare Sensor for Wearable Body Sensor Network

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Yan, Long; Yoo, Jerald; Kim, Binhee; Yoo, Hoi-Jun

被引用次数: 50

2010

A Low-Energy Inductive Coupling Transceiver With Cm-Range 50-Mbps Data Communication in Mobile Device Applications

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lee, Seulki; Song, Kiseok; Yoo, Jerald; Yoo, Hoi-Jun

被引用次数: 21

2010

A 118.4 GB/s Multi-Casting Network-on-Chip With Hierarchical Star-Ring Combined Topology for Real-Time Object Recognition

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Joo-Young; Park, Junyoung; Lee, Seungjin; Kim, Minsu; Oh, Jinwook; Yoo, Hoi-Jun

被引用次数: 14

2010

A 201.4 GOPS 496 mW Real-Time Multi-Object Recognition Processor With Bio-Inspired Neural Perception Engine

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Joo-Young; Kim, Minsu; Lee, Seungjin; Oh, Jinwook; Kim, Kwanho; Yoo, Hoi-Jun

被引用次数: 79

2010

A 5.2 mW Self-Configured Wearable Body Sensor Network Controller and a 12 μ W Wirelessly Powered Sensor for a Continuous Health Monitoring System

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Yoo, Jerald; Yan, Long; Lee, Seulki; Kim, Yongsang; Yoo, Hoi-Jun

被引用次数: 184

2010

An attention controlled multi-core architecture for energy efficient object recognition

SIGNAL PROCESSING-IMAGE COMMUNICATION

Kim, Joo-Young; Oh, Sejong; Lee, Seungjin; Kim, Minsu; Oh, Jinwook; Yoo, Hoi-Jun

被引用次数: 3

2010

A Low-Power Portable ECG Touch Sensor with Two Dry Metal Contact Electrodes

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Yan, Long; Yoo, Hoi-Jun

被引用次数: 14

2010

Selected Papers From the 2010 IEEE International Solid-State Circuits Conference (ISSCC)

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Hamoui, Anas A.; Yoo, Hoi-Jun

被引用次数: 0

2010

REAL-TIME OBJECT RECOGNITION WITH NEURO-FUZZY CONTROLLED WORKLOAD-AWARE TASK PIPELINING

IEEE MICRO

Kim, Joo-Young; Kim, Minsu; Lee, Seungjin; Oh, Jinwook; Oh, Sejong; Yoo, Hoi-Jun

被引用次数: 6

2009

A Low-Power Multimedia SoC with Fully Programmable 3D Graphics for Mobile Devices

IEEE COMPUTER GRAPHICS AND APPLICATIONS

Woo, Jeong-Ho; Kim, Hyejung; Yoo, Hoi-Jun; Sohn, Ju-Ho

被引用次数: 7

2009

Memory-centric network-on-chip for power efficient execution of task-level pipeline on a multi-core processor

IET COMPUTERS AND DIGITAL TECHNIQUES

Kim, D.; Kim, K.; Kim, J. -Y.; Lee, S.; Yoo, H. -J.

被引用次数: 8

2009

A Configurable Heterogeneous Multicore Architecture with Cellular Neural Network for Real-Time Object Recognition

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY

Kim, Kwanho; Lee, Seungjin; Kim, Joo-Young; Kim, Minsu; Yoo, Hoi-Jun

被引用次数: 12

2009

A 10.8 mW Body Channel Communication/MICS Dual-Band Transceiver for a Unified Body Sensor Network Controller

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Cho, Namjun; Bae, Joonsung; Yoo, Hoi-Jun

被引用次数: 93

2009

Introduction to the Special Section on the 2008 Asian Solid-State Circuits Conference (A-SSCC'08)

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Yoo, Hoi-Jun; Cho, Seong Hwan

被引用次数: 0

2009

A 1.12 pJ/b Inductive Transceiver With a Fault-Tolerant Network Switch for Multi-Layer Wearable Body Area Network Applications

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Yoo, Jerald; Lee, Seulki; Yoo, Hoi-Jun

被引用次数: 31

2009

An Embedded Stream Processor Core Based on Logarithmic Arithmetic for a Low-Power 3-D Graphics SoC

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Nam, Byeong-Gyu; Yoo, Hoi-Jun

被引用次数: 42

2009

A 60 kb/s-10 Mb/s Adaptive Frequency Hopping Transceiver for Interference-Resilient Body Channel Communication

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Cho, Namjun; Yan, Long; Bae, Joonsung; Yoo, Hoi-Jun

被引用次数: 109

2009

A 125 GOPS 583 mW Network-on-Chip Based Parallel Processor With Bio-Inspired Visual Attention Engine

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Kwanho; Lee, Seungjin; Kim, Joo-Young; Kim, Minsu; Yoo, Hoi-Jun

被引用次数: 30

2009

A 200-Mbps 0.02-nJ/b Dual-Mode Inductive Coupling Transceiver for cm-Range Multimedia Application

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Lee, Seulki; Yoo, Jerald; Yoo, Hoi-Jun

被引用次数: 14

2009

A Planar MICS Band Antenna Combined With a Body Channel Communication Electrode for Body Sensor Network

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Cho, Namjun; Roh, Taehwan; Bae, Joonsung; Yoo, Hoi-Jun

被引用次数: 56

2009

A 152-mW Mobile Multimedia SoC With Fully Programmable 3-D Graphics and MPEG4/H.264/JPEG

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Woo, Jeong-Ho; Sohn, Ju-Ho; Kim, Hyejung; Yoo, Hoi-Jun

被引用次数: 8	2009
81.6 GOPS Object Recognition Processor Based on a Memory-Centric NoC IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS Kim, Donghyun; Kim, Kwanho; Kim, Joo-Young; Lee, Seungjin; Lee, Se-Joong; Yoo, Hoi-Jun 被引用次数: 14	2009
Planar Fashionable Circuit Board Technology and Its Applications JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE Lee, Seulki; Kim, Binhee; Yoo, Hoi-Jun 被引用次数: 31	2009
Cost-effective low-power graphics processing unit for handheld devices IEEE COMMUNICATIONS MAGAZINE Nam, Byeong-Gyu; Lee, Jeabin; Kim, Kwanho; Lee, Seungjin; Yoo, Hoi-Jun 被引用次数: 6	2008
Power and area-efficient unified computation of vector and elementary functions for handheld 3D graphics systems IEEE TRANSACTIONS ON COMPUTERS Nam, Byeong-Gyu; Kim, Hyejung; Yoo, Hoi-Jun 被引用次数: 61	2008
A 195 mW, 9.1 Mvertices/s Fully Programmable. 3-D Graphics Processor for Low-Power Mobile Devices IEEE JOURNAL OF SOLID-STATE CIRCUITS Woo, Jeong-Ho; Sohn, Ju-Ho; Kim, Hyejung; Yoo, Hoi-Jun 被引用次数: 9	2008
A 195 mW/152 mW mobile multimedia SoC with fully programmable 3-D graphics and MPEG4/H.264/JPEG IEEE JOURNAL OF SOLID-STATE CIRCUITS Woo, Jeon-Ho; Sohn, Ju-Ho; Kim, Hyejung; Yoo, Hoi-Jun 被引用次数: 11	2008
A fully integrated digital hearing aid chip with human factors considerations IEEE JOURNAL OF SOLID-STATE CIRCUITS Kim, Sunyoung; Lee, Seung Jin; Cho, Namjun; Song, Seong-Jun; Yoo, Hoi-Jun 被引用次数: 22	2008
A 0.2-mW 2-mb/s digital transceiver based on wideband signaling for human body communications IEEE JOURNAL OF SOLID-STATE CIRCUITS Song, Seong-Jun; Cho, Namjun; Yoo, Hoi-Jun 被引用次数: 110	2007
A low-power unified arithmetic unit for programmable handheld 3-D graphics systems	

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Nam, Byeong-Gyu; Kim, Hyejung; Yoo, Hoi-Jun

被引用次数: 25

2007

The human body characteristics as a signal transmission medium for intrabody communication

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Cho, Namjun; Yoo, Jerald; Song, Seong-Jun; Lee, Jeabin; Jeon, Seonghyun; Yoo, Hoi-Jun

被引用次数: 282

2007

A 231-MHz, 2.18-mW 32-bit logarithmic arithmetic unit for fixed-point 3-D graphics system

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Hyejung; Nam, Byeong-Gyu; Sohn, Ju-Ho; Woo, Jeong-Ho; Yoo, Hoi-Jun

被引用次数: 70

2006

An autonomous SRAM with on-chip sensors in an 80-nm double stacked cell technology

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Sohn, K; Mo, HS; Suh, YH; Byun, HG; Yoo, HJ

被引用次数: 6

2006

An energy-efficient analog front-end circuit for a sub-1-V digital hearing aid chip

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, S; Lee, JY; Song, SJ; Cho, NJ; Yoo, HJ

被引用次数: 64

2006

A regulated charge pump with small ripple voltage and fast start-up

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lee, JY; Kim, SE; Song, SJ; Kim, JK; Kim, S; Yoo, HJ

被引用次数: 86

2006

Chip-package hybrid clock distribution network and DLL for low jitter clock delivery

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Chung, D; Ryu, C; Kim, H; Lee, C; Kim, J; Bae, K; Yu, J; Yoo, H; Kim, J

被引用次数: 10

2006

Low-power network-on-chip for high-performance SoC design

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Lee, KM; Lee, SJ; Yoo, HJ

被引用次数: 127

2006

Low-power 3D graphics processors for mobile terminals

IEEE COMMUNICATIONS MAGAZINE

Sohn, JH; Park, YH; Yoon, CW; Woo, R; Park, SJ; Yoo, HJ

被引用次数: 12

2005

A 0.7-fJ/bit/search 2.2-ns search time hybrid-type TCAM architecture

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Choi, S; Sohn, K; Yoo, HJ

被引用次数: 52

2005

Packet-switched on-chip interconnection network for system-on-chip applications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Lee, SJ; Lee, KM; Song, SJ; Yoo, HJ

被引用次数: 31

2005

A 210-mW graphics LSI implementing full 3-D pipeline with 264 Mtexels/s texturing for mobile multimedia applications

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Woo, R; Choi, S; Sohn, JH; Song, SJ; Yoo, HJ

被引用次数: 18

2004

1.25-Gb/s regulated cascode CMOS transimpedance amplifier for Gigabit Ethernet applications

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Park, SM; Yoo, HJ

被引用次数: 284

2004

A 4-gb/s CMOS clock and data recovery circuit using 1/8-rate clock technique

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Song, SJ; Park, SM; Yoo, HJ

被引用次数: 60

2003

Low power motion estimation and motion compensation block IPs in MPEG-4 video codec hardware for portable applications

IEICE TRANSACTIONS ON ELECTRONICS

Yoon, CW; Yoo, HJ

被引用次数: 3

2003

A reconfigurable multilevel parallel texture cache memory with 75-GB/s parallel cache replacement bandwidth

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Park, SJ; Kim, JS; Woo, R; Lee, SJ; Lee, KM; Yang, TH; Jung, JY; Yoo, HJ

被引用次数: 3

2002

Race Logic Architecture (RALA): A novel logic concept using the race scheme of input variables

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lee, SJ; Yoo, HJ

被引用次数: 7

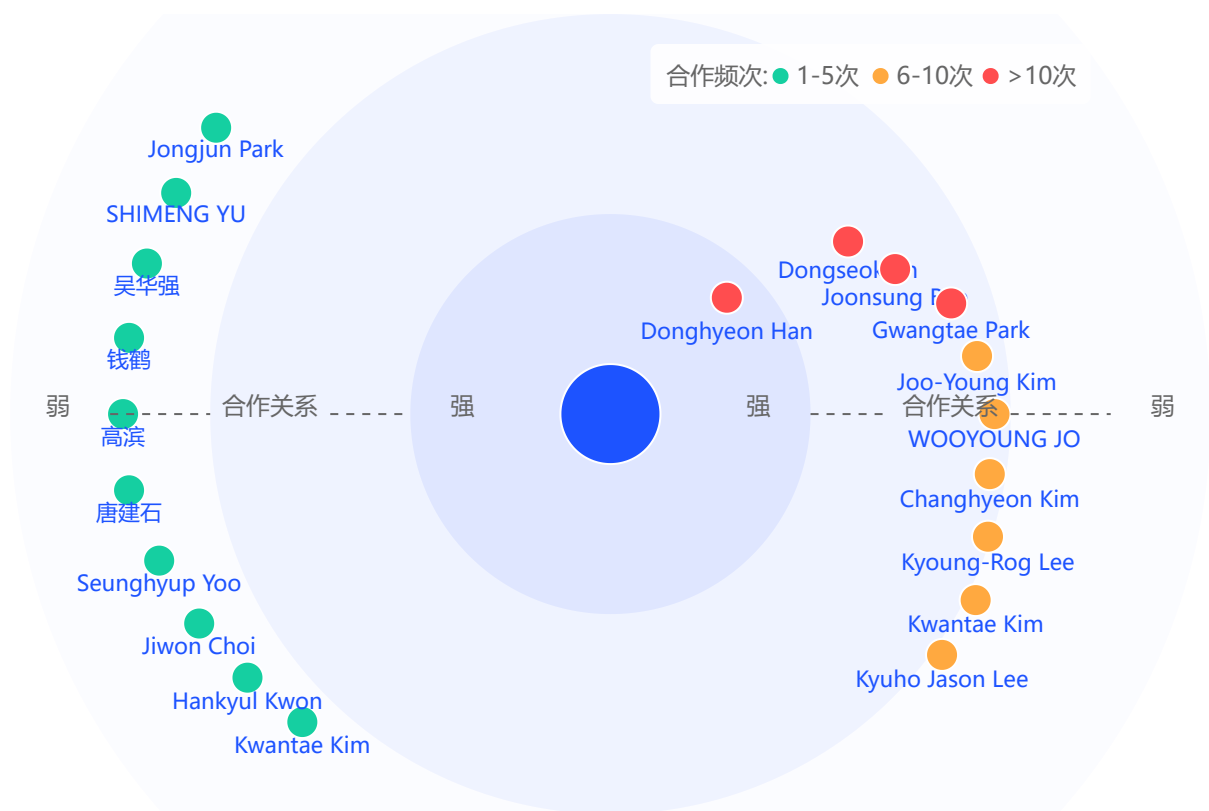
2002

Bit-wise read-compare-write scheme for low power read-modify-write DRAM operation

ELECTRONICS LETTERS

Park, YH; Choi, S; Yoo, HJ	
被引用次数: 5	2002
Embedded DRAM (eDRAM) power-energy estimation using signal swing-based analytical model	
IEICE TRANSACTIONS ON ELECTRONICS	
Park, YH; Kook, J; Yoo, HJ	
被引用次数: 0	2002
A 7.1-GB/s low-power rendering engine in 2-D array-embedded memory logic CMOS for portable multimedia system	
IEEE JOURNAL OF SOLID-STATE CIRCUITS	
Park, YH; Han, SH; Lee, JH; Yoo, HJ	
被引用次数: 15	2001
Hidden double data transfer scheme for MDL design	
ELECTRONICS LETTERS	
Park, SJ; Yoo, HJ	
被引用次数: 0	2001
Design of a temperature-compensated crystal oscillator using the new digital trimming method	
JOURNAL OF THE KOREAN PHYSICAL SOCIETY	
Je, M; Lee, K; Gil, J; Yoo, HJ; Shin, I	
被引用次数: 0	2000
The Heterogeneous Deep Neural Network Processor With a Non-von Neumann Architecture	
Proc.IEEE	
Dongjoo Shin; Hoi-Jun Yoo	
被引用次数: 0	

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